Practice workbook · Variables, constants, arrays and records

Definitions, purposes and benefits of variables and constants, and how arrays (1D/2D) and records differ from single variables and from each other.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-02

1. Question 1 · part c.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.ii only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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1. Question 1 · part c.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.ii only.

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(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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1. Question 1 · part c.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.ii only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Identify the name of a variable that holds a real number.

(1)

(iv) Identify the name of a parameter.

(1)

(Total for Question 1 = 14 marks)

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Working and answer

2. Question 1 · part d · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part d only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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2

P 6 1 8 8 5 A 0 2 1 6

2. Question 1 · part d · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part d only.

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(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

.....

.....

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

.....

(ii) Identify the line number where a subprogram definition starts.

(1)

.....

(iii) Identify the name of a numeric variable.

(1)

.....

.....

(iv) Identify the name of a string variable.

(1)

.....

.....

(Total for Question 1 = 10 marks)

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2. Question 1 · part d · 1 mark · Working and answer

Working and answer

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3. Question 1 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

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3. Question 1 · part b · 2 marks · Working and answer

Working and answer

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4. Question 1 · part a · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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4. Question 1 · part a · 1 mark · Working and answer

Working and answer

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5. Question 1 · part c.i · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c.i only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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5. Question 1 · part c.i · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c.i only.

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(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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5. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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6. Question 1 · part c.ii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c.ii only.

1 Programs are used to handle financial transactions.

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Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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6. Question 1 · part c.ii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c.ii only.

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(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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6. Question 1 · part c.ii · 1 mark · Working and answer

Working and answer

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7. Question 1 · part d · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part d only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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7. Question 1 · part d · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part d only.

(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

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7. Question 1 · part d · 1 mark · Working and answer

Working and answer

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8. Question 2 · part a.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a.ii only.

2 Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

- ☒ **A** Algorithm
- ☒ **B** Computational thinking
- ☒ **C** Decomposition
- ☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

.....

.....

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

.....

.....

(ii) Give **one** example of a runtime error.

(1)

.....

.....

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8. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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9. Question 1 · part c.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part c.i only.

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(1)

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(b) Computer code can contain errors.

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Description	Syntax error	Logic error	Runtime error
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9. Question 1 · part c.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part c.i only.

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(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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9. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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